

SIMATS ENGINEERING

CO3-ASSESSMENT TOOL1- CASE STUDY

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:40

Code of Conduct:

*I __Shaik Abdul Razak (192524358)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: *Shaik Abdul Razak*

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- II. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- III. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) $P1: \text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- 2) $P2: \text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
- 3) $P3: \neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: $\text{SoilDry} = \text{True}, \text{CropWheat} = \text{True}$

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) $\text{Stench } [1,2] \rightarrow \text{Wumpus is adjacent to cell } [1,2]$
- 2) $\text{Breeze } [1,1] \rightarrow \text{Pit is adjacent to cell } [1,1]$
- 3) $\text{Glitter } [x,y] \rightarrow \text{Gold is at } [x,y]$
- 4) $\neg \text{Wumpus } [x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

CASE STUDY 1 — SMART MEDICAL DIAGNOSIS SYSTEM

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

1. If a patient has Fever AND Cough $\rightarrow$ Possible Flu.
2. If a patient has Fever AND Rash $\rightarrow$ Possible Measles.
3. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia.
4. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True.

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

Let the following propositional symbols represent the facts:

F = Patient A has Fever.

C = Patient A has Cough.

R = Patient A has Rash.

B = Patient A has Breathlessness.

FL = Patient A has Possible Flu.

M = Patient A has Possible Measles.

P = Patient A has Possible Pneumonia.

The given rules can be represented as follows:

Rule 1:

$F \wedge C \rightarrow FL$

This means: If Patient A has Fever AND Cough, then Patient A has Possible Flu.

Rule 2:

$F \wedge R \rightarrow M$

This means: If Patient A has Fever AND Rash, then Patient A has Possible Measles.

Rule 3:

$C \wedge B \rightarrow P$

This means: If Patient A has Cough AND Breathlessness, then Patient A has Possible Pneumonia.

Given facts:

F = True

C = True

B = True

Therefore, the Knowledge Base is:

$KB = \{F, C, B, F \wedge C \rightarrow FL, F \wedge R \rightarrow M, C \wedge B \rightarrow P\}$

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient

A. Show each inference step with the rule fired and fact derived.

Forward Chaining is a data-driven inference method. It starts with the known facts and applies rules whose conditions are satisfied.

Initial facts:

F = True

C = True

B = True

Step 1: Apply Rule 1

Rule 1:

$F \wedge C \rightarrow FL$

Since:

F = True

C = True

Both conditions are satisfied.

Therefore:

FL = True

Hence:

Possible Flu is derived.

Step 2: Apply Rule 2

Rule 2:

$F \wedge R \rightarrow M$

We know:

F = True

But R = Rash is not given as a fact.

Therefore, Rule 2 cannot be fired.

Hence, Possible Measles cannot be derived.

Step 3: Apply Rule 3

Rule 3:

$C \wedge B \rightarrow P$

Since:

C = True

B = True

Both conditions are satisfied.

Therefore:

P = True

Hence:

Possible Pneumonia is derived.

Final Result of Forward Chaining

The diagnoses that can be derived for Patient A are:

Possible Flu = True

Possible Pneumonia = True

Possible Measles cannot be derived because Rash is not known to be true.

Therefore:

Final Answer:

Patient A has Possible Flu and Possible Pneumonia according to the given rules.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Goal:

Possible Pneumonia(A)

Backward Chaining starts from the goal and searches for a rule that can produce the goal.

Rule 3 is:

$Cough(A) \wedge Breathlessness(A) \rightarrow Possible\ Pneumonia(A)$

Therefore, the goal is reduced into two subgoals:

Cough(A)

and

Breathlessness(A)

Both are given as facts.

Therefore:

Cough(A) = True

Breathlessness(A) = True

Hence:

Possible Pneumonia(A) = True

Goal Reduction Tree

Goal: **Possible Pneumonia(A)**
 Possible Pneumonia(A)

Therefore, the goal is successfully verified.

Since **Cough(A)** and **Breathlessness(A)** are both known facts, Rule III can be applied:

Cough(A) $\wedge$ Breathlessness(A) $\rightarrow$ Possible Pneumonia(A)

Therefore:

Patient A has **Possible Pneumonia**.

CASE STUDY 2 — AUTONOMOUS TRAFFIC MANAGEMENT AGENT

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing.

Given rules:

Rule 1:

$\forall x: \text{Vehicle}(x) \wedge \text{Emergency Type}(x) \rightarrow \text{ClearPath}(x)$

Rule 2:

$\forall x: \text{ClearPath}(x) \rightarrow \text{Green Signal}(x) \wedge \neg \text{Red Signal}(x)$

Rule 3:

$\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x, y) \wedge \text{Green Signal}(y) \rightarrow \text{Proceed}(x)$

Given facts:

Vehicle (Ambulance)

Emergency Type (Ambulance)

Vehicle (Car A)

Behind (Car A, Ambulance)

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

Rule 1 is:

$\forall x: \text{Vehicle}(x) \wedge \text{Emergency Type}(x) \rightarrow \text{ClearPath}(x)$

The relevant facts are:

Vehicle (Ambulance)

Emergency Type (Ambulance)

Step 1

Match:

Vehicle(x)

with:

Vehicle (Ambulance)

Therefore:

$\theta_1 = \{x/\text{Ambulance}\}$

Step 2

Match:

Emergency Type(x)

with:

Emergency Type(Ambulance)

Therefore:

$\theta_2 = \{x/\text{Ambulance}\}$

Both substitutions are consistent.

Therefore, the Most General Unifier is:

$\theta = \{x/\text{Ambulance}\}$

Applying the substitution to Rule 1:

Vehicle (Ambulance) $\wedge$ Emergency Type (Ambulance) $\rightarrow$ Clear Path (Ambulance)

Since both facts are true:

ClearPath (Ambulance)

is derived.

Final Answer:

MGU:

$\theta = \{x/\text{Ambulance}\}$

Derived fact:

Clear Path (Ambulance)

(b) Using the Resolution method, prove that 'Proceed (Car A)' is true. Convert relevant clauses to CNF and show each resolution step.

Goal:

Proceed (Car A)

To prove the goal using Resolution Refutation, we add the negation of the goal:

Proceed (Car A)

Step 1: Convert Rule 1 to CNF

Original Rule 1:

Vehicle(x) $\wedge$ Emergency Type(x) $\rightarrow$ ClearPath(x)

Using:

$A \rightarrow B \equiv A \vee B$

we get:

Vehicle(x) $\vee$ Emergency Type(x) $\vee$ ClearPath(x)

Therefore:

C1 = Vehicle(x) $\vee$ Emergency Type(x) $\vee$ ClearPath(x)

Step 2: Convert Rule 2 to CNF

Original Rule 2:

ClearPath(x) $\rightarrow$ Green Signal(x) $\wedge$ Red Signal(x)

This gives two clauses:

C2 = ClearPath(x) $\vee$ Green Signal(x)

C3 = ClearPath(x) $\vee$ Red Signal(x)

Step 3: Convert Rule 3 to CNF

Original Rule 3:

Vehicle(x) $\wedge$ Behind (x, y) $\wedge$ Green Signal(y) $\rightarrow$ Proceed(x)

Therefore:

C4 = Vehicle(x) $\vee$ Behind (x, y) $\vee$ Green Signal(y) $\vee$ Proceed(x)

Facts

C5 = Vehicle (Ambulance)

C6 = Emergency Type (Ambulance)

C7 = Vehicle (Car A)

C8 = Behind (Car A, Ambulance)

Negated Goal

C9 = Proceed (Car A)

Resolution Step 1

Use C1:

Vehicle(x) $\vee$ Emergency Type(x) $\vee$ ClearPath(x)

and C5:

Vehicle (Ambulance)

Substitute:

x = Ambulance

Resolving gives:

Emergency Type (Ambulance) $\vee$ Clear Path (Ambulance)

Now resolve with C6:

Emergency Type (Ambulance)

Therefore:

$C10 = \text{Clear Path (Ambulance)}$

Resolution Step 2

Use C2:

$\text{ClearPath}(x) \vee \text{Green Signal}(x)$

and C10:

$\text{Clear Path (Ambulance)}$

Substitute:

$x = \text{Ambulance}$

Therefore:

$C11 = \text{Green Signal (Ambulance)}$

Resolution Step 3

Use C4:

$\text{Vehicle}(x) \vee \text{Behind}(x, y) \vee \text{Green Signal}(y) \vee \text{Proceed}(x)$

Substitute:

$x = \text{Car A}$

$y = \text{Ambulance}$

Therefore:

$\text{Vehicle (Car A)} \vee \text{Behind (Car A, Ambulance)} \vee \text{Green Signal (Ambulance)} \vee \text{Proceed (Car A)}$

Using:

Vehicle (Car A)

$\text{Behind (Car A, Ambulance)}$

$\text{Green Signal (Ambulance)}$

we derive:

$C12 = \text{Proceed (Car A)}$

Resolution Step 4

We have:

Proceed (Car A)

and the negated goal:

Proceed (Car A)

Resolving these gives:

The empty clause indicates a contradiction.

Therefore:

Proceed (Car A) is proven.

Resolution Proof Chain

$\text{Vehicle (Ambulance)} + \text{Emergency Type (Ambulance)}$

↓

$\text{Clear Path (Ambulance)}$

↓

$\text{Green Signal (Ambulance)}$

↓

$\text{Vehicle (Car A)} + \text{Behind (Car A, Ambulance)} + \text{Green Signal (Ambulance)}$

↓

Proceed (Car A)

↓

$\text{Proceed (Car A)} + \text{Proceed (Car A)}$

↓

Final Conclusion:

Proceed (Car A) is true.

(c) Analyse whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

The current knowledge base can identify emergency vehicles individually.

For example, if there are two emergency vehicles:

Vehicle (Ambulance1)

Emergency Type (Ambulance1)

Vehicle (Ambulance2)

Emergency Type (Ambulance2)

Rule 1 can derive:

ClearPath (Ambulance1)

ClearPath (Ambulance2)

However, the current knowledge base does not contain rules for resolving conflicts when two emergency vehicles require the same road or intersection at the same time.

Additional information is required, such as:

1. Location of each emergency vehicle.
2. Direction of movement of each vehicle.
3. Priority of one emergency vehicle over another.
4. Road or intersection required by each vehicle.
5. Rules for handling conflicting routes.

For example, an additional rule could be:

$\text{Emergency Priority}(x, y) \wedge \text{Emergency Type}(x) \rightarrow \text{Give Priority}(x)$

Another rule could be:

$\text{Priority}(x, y) \wedge \text{Requires Route}(x, r) \wedge \text{Requires Route}(y, r) \rightarrow \text{Allow First}(x)$

Therefore, the current knowledge base is sufficient for basic emergency vehicle routing, but it is not sufficient for handling two simultaneous emergency vehicles when their routes conflict.

CASE STUDY 3 — AGRICULTURAL EXPERT REASONING SYSTEM

The agricultural advisory logical agent has the following propositional knowledge base:

P1: Soil Dry $\rightarrow$ Irrigation Needed

P2: Irrigation Needed $\wedge$ Crop Wheat $\rightarrow$ Apply Drip Method

P3: Apply Drip Method $\wedge$ Crop Wheat $\rightarrow$ Crop at Risk

Facts:

Soil Dry = True

Crop Wheat = True

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step.

P1

Original:

Soil Dry $\rightarrow$ Irrigation Needed

Eliminate implication:

soil Dry $\vee$ Irrigation Needed

Therefore:

CNF(P1) = Soil Dry $\vee$ Irrigation Needed

P2

Original:

Irrigation Needed $\wedge$ Crop Wheat $\rightarrow$ Apply Drip Method

Eliminate implication:

(Irrigation Needed $\wedge$ Crop Wheat) $\vee$ Apply Drip Method

Using De Morgan's Law:

Irrigation Needed $\vee$ Crop Wheat $\vee$ Apply Drip Method

Therefore:

CNF(P2) = Irrigation Needed $\vee$ Crop Wheat $\vee$ Apply Drip Method

P3

Original:

Apply Drip Method $\wedge$ Crop Wheat $\rightarrow$ Crop at Risk

Eliminate implication:

(Apply Drip Method $\wedge$ Crop Wheat) $\vee$ Crop at Risk

Using De Morgan's Law:

Apply Drip Method $\vee$ Crop Wheat $\vee$ Crop at Risk

Therefore:

CNF(P3) = Apply Drip Method $\vee$ Crop Wheat $\vee$ Crop at Risk

Facts

Soil Dry = True

CNF:

Soil Dry

Crop Wheat = True

CNF:

Crop Wheat

Final CNF Knowledge Base

C1 = Soil Dry $\vee$ Irrigation Needed

C2 = Irrigation Needed $\vee$ $\neg$ Crop Wheat $\vee$ Apply Drip Method

C3 = Apply Drip Method $\vee$ Crop Wheat $\vee$ Crop at Risk

C4 = Soil Dry

C5 = Crop Wheat

(b) Using Resolution Refutation, prove the theorem ‘Apply Drip Method’. Negate the goal, combine with the KB in CNF, and derive the empty clause. Draw the resolution proof tree.

Goal:

Apply Drip Method

Negate the goal:

Apply Drip Method

Add Apply Drip Method to the CNF knowledge base.

Resolution Step 1

Use:

C1 = Soil Dry $\vee$ Irrigation Needed

and:

C4 = Soil Dry

Resolving them gives:

Irrigation Needed

Therefore:

C6 = Irrigation Needed

Resolution Step 2

Use:

C2 = Irrigation Needed $\vee$ Crop Wheat $\vee$ Apply Drip Method

and:

C6 = Irrigation Needed

Resolving gives:

Crop Wheat $\vee$ Apply Drip Method

Therefore:

C7 = Crop Wheat $\vee$ Apply Drip Method

Resolution Step 3

Use:

C5 = Crop Wheat

and:

C7 = Crop Wheat $\vee$ Apply Drip Method

Resolving gives:

Apply Drip Method

Therefore:

C8 = Apply Drip Method

Resolution Step 4

We have:

Apply Drip Method

and the negated goal:

Apply Drip Method

Resolving them gives:

The empty clause indicates a contradiction.

Therefore, the original goal is proven.

Final Answer:

Apply Drip Method is true.

Therefore:

Apply Drip Method is proven by Resolution Refutation.

(c) If the fact 'Soil Dry' is removed from the KB, analyse step-by-step how logical conclusions change. Does 'Crop at Risk' still follow?

Suppose the fact:

Soil Dry

is removed.

The remaining fact is:

Crop Wheat

Step 1

P1 is:

Soil Dry $\rightarrow$ Irrigation Needed

Since Soil Dry is no longer known to be true, Irrigation Needed cannot be derived.

Therefore:

Irrigation Needed cannot be concluded.

Step 2

P2 is:

Irrigation Needed $\wedge$ Crop Wheat $\rightarrow$ Apply Drip Method

Although Crop Wheat is true, Irrigation Needed is not known.

Therefore:

Apply Drip Method cannot be derived.

Step 3

P3 is:

Apply Drip Method $\wedge$ Crop Wheat $\rightarrow$ Crop At Risk

We know:

Crop Wheat = True

But we do not know:

Apply Drip Method

The absence of Apply Drip Method does not mean that Apply Drip Method is true.

Therefore:

Crop At Risk cannot be derived.

Final Conclusion

After removing Soil Dry:

Irrigation Needed cannot be derived.

Apply Drip Method cannot be derived.

Crop At Risk cannot be derived.

Therefore:

No, Crop At Risk does not logically follow from the remaining knowledge base.

This is because the system follows the principle that failure to prove a statement does not mean its negation is true.

CASE STUDY 4 — WUMPUS WORLD KNOWLEDGE AGENT

An AI agent navigating the Wumpus World perceives:

Stench at [1,2]

Breeze at [1,1]

Glitter at [2,2]

The agent uses the following propositional knowledge base:

Rule 1:

$\text{Stench}[1,2] \rightarrow \text{Wumpus is adjacent to cell}[1,2]$

Rule 2:

$\text{Breeze}[1,1] \rightarrow \text{Pit is adjacent to cell}[1,1]$

Rule 3:

$\text{Glitter}[x, y] \rightarrow \text{Gold is at}[x, y]$

Rule 4:

$\text{Wumpus}[x, y] \wedge \neg \text{Pit}[x, y] \rightarrow \text{Safe}[x, y]$

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

Given Percepts

The agent perceives:

Stench [1,2]

Breeze [1,1]

Glitter [2,2]

Rule 1

$\text{Stench}[1,2] \rightarrow \text{Wumpus Adjacent}[1,2]$

Since:

Stench [1,2]

is true, by Modus Ponens:

Wumpus Adjacent [1,2]

is derived.

Therefore:

Wumpus is adjacent to [1,2].

Rule 2

$\text{Breeze}[1,1] \rightarrow \text{Pit Adjacent}[1,1]$

Since:

Breeze [1,1]

is true, by Modus Ponens:

Pit Adjacent [1,1]

is derived.

Therefore:

A Pit is adjacent to [1,1].

Rule 3

$\text{Glitter}[x, y] \rightarrow \text{Gold}[x, y]$

Given:

Glitter [2,2]

Substitute:

$x = 2$

$y = 2$

Therefore:

Gold [2,2]

is derived.

Rule 4

$\text{Wumpus}[x, y] \wedge \neg \text{Pit}[x, y] \rightarrow \text{Safe}[x, y]$

There are no explicit facts proving that a particular cell does not contain a Wumpus or Pit. Therefore, Rule 4 cannot be fired with the given information.

Derived Conclusions

1. Stench[1,2]
2. Breeze[1,1]
3. Glitter[2,2]
4. WumpusAdjacent[1,2]
5. PitAdjacent[1,1]
6. Gold[2,2]

Therefore, the agent knows that gold is located at [2,2], but it does not have enough information to declare the surrounding cells completely safe.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

Forward Chaining begins with the percepts and derives possible locations.

Step 1: Stench at [1,2]

Given:

Stench[1,2]

Rule:

Stench[1,2] $\rightarrow$ Wumpus is adjacent to [1,2]

In a standard four-directional grid, the cells adjacent to [1,2] are:

[1,1]

[1,3]

[2,2]

Therefore, the possible Wumpus locations are:

Wumpus $\in \{[1,1], [1,3], [2,2]\}$

Step 2: Breeze at [1,1]

Given:

Breeze[1,1]

Rule:

Breeze[1,1] $\rightarrow$ Pit is adjacent to [1,1]

The adjacent cells are:

[1,2]

[2,1]

Therefore, the possible Pit locations are:

Pit $\in \{[1,2], [2,1]\}$

Step 3: Glitter at [2,2]

Given:

Glitter[2,2]

Rule:

Glitter[x,y] $\rightarrow$ Gold[x,y]

Therefore:

Gold[2,2]

Hence, the gold is located at:

[2,2]

Final Possible Locations

Possible Wumpus locations:

[1,1], [1,3], [2,2]

Possible Pit locations:

[1,2], [2,1]

Gold location:

[2,2]

(c) Evaluate whether the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

The proposed path is:

$[1,1] \rightarrow [1,2] \rightarrow [2,2]$

Cell [1,1]

The agent perceives:

Breeze[1,1]

This means that a Pit may be present in a neighboring cell.

The possible Pit locations are:

[1,2] and [2,1]

Therefore, the agent cannot conclude that [1,2] is safe.

Cell [1,2]

The agent perceives:

Stench[1,2]

This indicates that a Wumpus may be in an adjacent cell.

Also, from the Breeze at [1,1], a Pit may be located at [1,2].

Therefore, [1,2] cannot be guaranteed safe.

Cell [2,2]

The agent knows:

Gold[2,2]

Therefore, [2,2] contains the gold.

However, from Stench[1,2], [2,2] is one of the possible Wumpus locations.

Therefore, the agent cannot guarantee that [2,2] is safe.

Safety Rule

The safety rule is:

$\text{Wumpus}[x,y] \wedge \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

To prove that a cell is safe, the agent needs both:

$\text{Wumpus}[x,y]$

and

$\text{Pit}[x,y]$

These negative facts are not available for all cells on the proposed path.

Therefore, the complete path cannot be logically guaranteed to be safe.

Final Conclusion

The path:

$[1,1] \rightarrow [1,2] \rightarrow [2,2]$

is NOT guaranteed to be safe based on the available knowledge.

The agent should not assume that the path is safe because:

1. [1,2] may contain a Pit.
2. [2,2] may contain a Wumpus according to the available Stench information.
3. The KB does not provide sufficient negative facts to prove the cells are safe.

Therefore:

Final Answer:

The path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ cannot be confirmed as safe with the given knowledge base. More percepts or additional knowledge are required before the agent can safely move through the path.

